



Enhancing Systems Engineering Academic Processes with AI at NPS: Initial Experiences

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Motivation

- Systems Engineering graduate education depends on rigorous but labor-intensive academic processes.
- Education technician support, advising, thesis/capstone assessment, grading, and accreditation evidence require consistency and traceability.
- Staff and faculty time are constrained, while institutional and defense-relevance expectations remain high.
- The SE Department is using governed AI assistants as early adopters for practical campus implementation.
- The goal is not autonomous decision making, but better human-reviewed draft products, evidence, feedback, and process continuity.

Current SE Department AI Assistant Portfolio

- **HEATHER 2:** supports education technician and academic administration processes using SOPs and policy documents.
- **Capstone/Thesis Reviewer:** generates rubric-based draft reviews and structured assessment outputs for faculty review.
- **Automatic Grader:** course-level grading assistant under development for assignment assessment and formative feedback.
- **AI Bootstrapper:** helps instantiate new governed assistants from reusable instruction, knowledge-base, and testing patterns.
- These efforts are being executed locally in SE while establishing reusable patterns for broader NPS adoption.

HEATHER 2: Education Technician Support

- Custom GPT grounded in SE Department SOPs, cheat sheets, and position guidance.
- Supports matriculation, ABET checks, enrollment management, matrices, graduation, and routine communications.
- Provides just-in-time support for new personnel and recurring administrative cases.
- Formal configuration management board is now used for knowledge-base and instruction updates.
- Human users retain responsibility for official actions and student records.

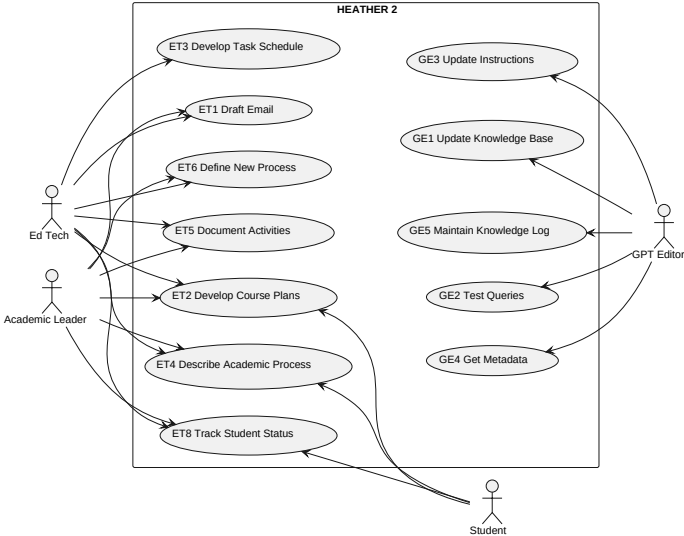
HEATHER 2: Education Technician Support

- Custom GPT grounded in SE Department Standard Operating Procedures (SOPs).
- Supports matriculation, ABET checks, enrollment management, student matrices, graduation, and routine communications.
- In routine daily usage by Ed Tech and faculty.



Illustration by HEATHER 2

Representative HEATHER 2 Use Cases



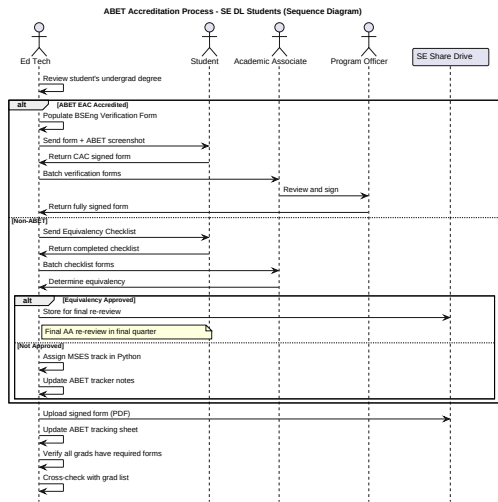
Current Documents in HEATHER 2 Knowledge Base

Document	Summary
Business Continuity Plan for SE (DL) Operations	Procedures to sustain DL academic operations during outages or disruptions.
ABET SOP for MSSE Program	Workflow for ABET/equivalency verification to determine MSSE degree eligibility.
Graduation Process SOP	Timelines, forms, and approval chains for SE DL degree and certificate completion.
New Student Enrollment Cheat Sheets	Procedures for Python matrix assignment, track designation, and academic commenting.
Matrix Management Cheat Sheets	Instructions for editing matrices, verifying certificate eligibility, and updating EDD/EGD.
Catalog Update Procedures	Step-by-step instructions for updating SmartCatalog entries for SE curricula.
Applicant Reporting Cheat Sheets	Filters, date ranges, and formatting guidelines for generating applicant data from Python.
INCOSE Certification Support SOP	Eligibility criteria and communication workflow for ASEP/CSEP via SE3100 equivalency.
Outstanding Thesis Selection SOP	End-to-end nomination, committee, and approval workflow for Outstanding Thesis awards.
Administrative and Academic Process Guides	High-level guidance on recurring responsibilities, communications, and records practices.
Education Technician Position Description (GS-1702-09)	Federal classification guidance on core Ed Tech responsibilities, authority, and evaluation.
heather.py	Current usage model for the GPT with defined use cases, associated testing data, and chat history. This is for self awareness, auditing and reporting

Example Ed Tech Use Case Testing Results

ID	Use Case Name	Tester(s)	Date	Result	Test Session Links
ET1	Draft Email	Madachy	2025-04-22	Pass	Student Email
ET1	Draft Email	Madachy	2025-04-22	Pass	Cohort Email
ET1	Draft Email	Samaha	2025-04-29	Pass	Draft Sponsor Emails
ET2	Develop Course Plans	Group	2025-04-25	Pass	Standard Course Plan
ET2	Develop Course Plans	Bell	2025-04-26	Pass	PhD Remediation Plan
ET2.1	Develop Custom Course Plans	Group	2025-04-25	Pass	Customized Course Plan
ET3	Develop Task Schedule	Madachy	2025-04-22	Pass	Graduation Task Plan
ET3	Develop Task Schedule	Samaha	2025-05-02	Pass	Ed Tech Calendar Analysis
ET4	Describe Academic Process	Bell	2025-04-26	Pass	Candidacy Steps
ET4	Describe Academic Process	Bell	2025-04-26	Pass	PhD Exams
ET4	Describe Academic Process	Madachy	2025-04-22	Pass	Draw ABET Accreditation Sequence
ET5	Document Activities	Samaha	2025-04-29	Pass	Progress Reporting
ET6	Define New Process	Madachy	2025-04-28	Pass	Document process improvement

HEATHER 2 Example: Describe Academic Process



Capstone/Thesis Reviewer

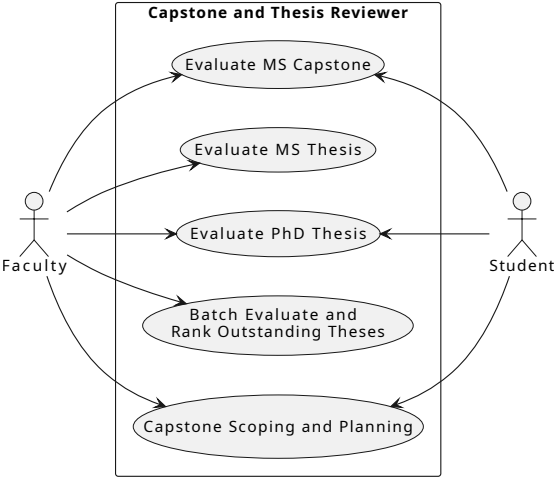
- Faculty-facing assistant for draft assessment of theses and capstone reports.
- Applies NPS learning outcomes, SE rubrics, defense relevance, technical rigor, and writing criteria.
- Produces coordinated narrative reviews and SpreadsheetML assessment workbooks.
- Current work is refining instructions, calibration, and experiments against faculty judgments and golden examples.
- Results support student feedback, accreditation evidence, and cohort-level analysis.

Placeholder
for use case
diagram

figures/capstone

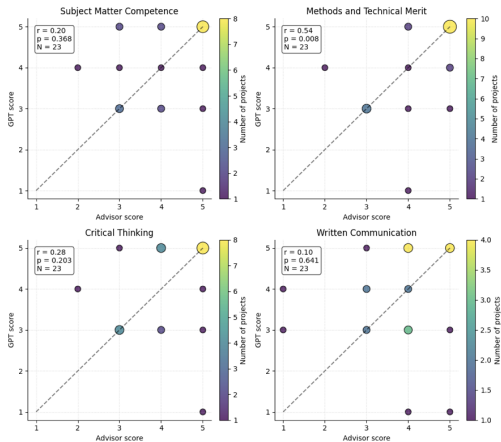
Capstone/Thesis
Reviewer
use cases

Demonstrated Capstone/Thesis Reviewer Use Cases

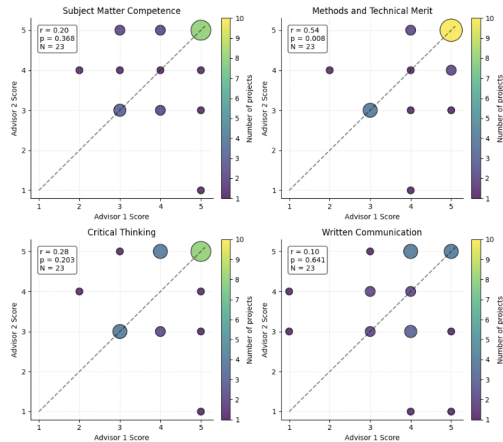


Towards Assessment Calibration

Advisor vs GPT Scores by Learning Outcome
Bubble size and color = number of projects



Advisor 1 vs Advisor 2 Scores by Learning Outcome
Bubble size and color = number of reports



Automatic Grader

- Course-level grading assistant under development for SE instructional use.
- Applies assignment-specific rubrics, identifies missing reasoning, and drafts formative feedback.
- Initial experiments compare AI-generated feedback with instructor grading and student revision outcomes.
- Planned outputs include a review narrative, rubric scores, common-error summaries, and machine-readable records.
- Final grades remain instructor decisions; student disclosure and course policy language are part of

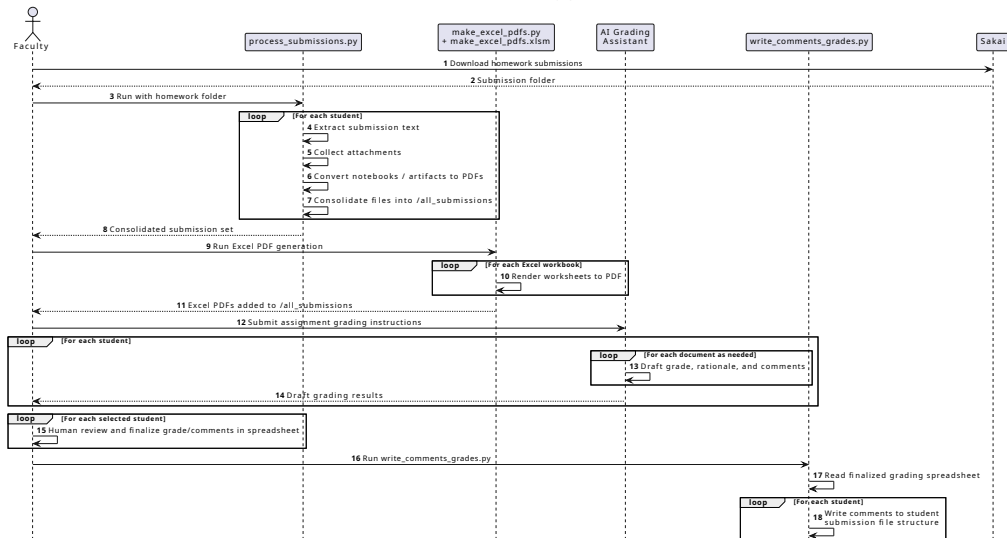
Placeholder
for use case
diagram

figures/automatic

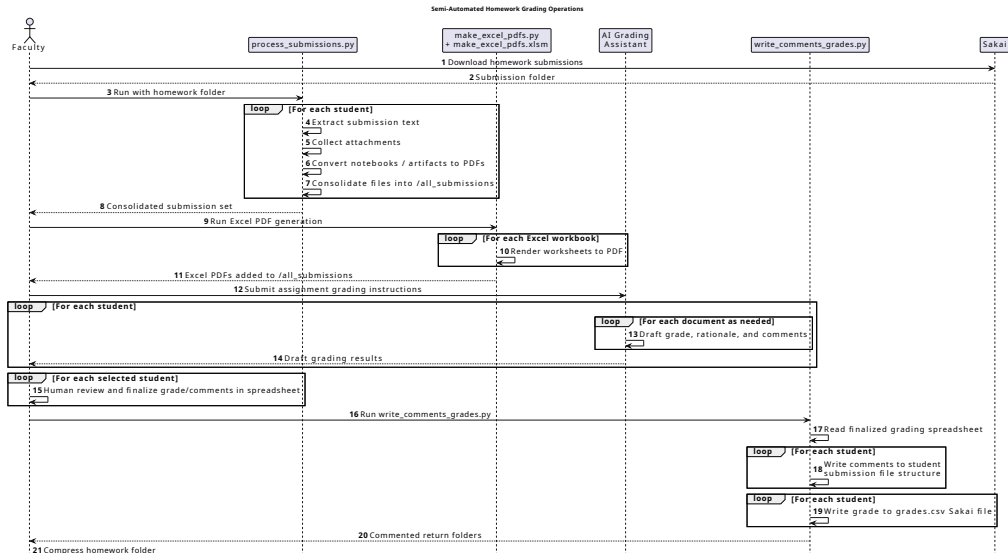
Automatic
Grader use
cases

Grading Sequence

Semi-Automated Homework Grading Operations



Grading Sequence



Automated Grader: Processing Constraints

- Student submissions are accepted in any format, so grading starts with artifact normalization rather than AI review.
- Most submitted artifacts are converted to PDF for reliable AI parsing and consistent visual review.
- Text-native files such as `.txt` and `.md` can be used directly without conversion.
- Colab URLs, `.ipynb` notebooks, Excel files, and other attachments are converted into reviewable PDF artifacts.
- Excel requires special handling: embedded graphics, charts, text boxes, formatting, and multi-tab layouts may not be readable from cell values alone.
- The Excel macro is used to render each workbook faithfully through Excel's own rendering engine; pandas-based extraction is not sufficient for these cases.

Example AI Grader Output

Student Name	Submission Text	#5 Effort	#5 Schedule	#5 Cost	#5 Avg Staffing	...	Total Points	Comments
ANSWER KEY	—	145.1 PM	7.8 Months	\$1,741,761	18.6 People	...	—	—
Doe, John	—	225 PM	9 Months	\$2,700,261	25 People	...	19.0	John Doe (19/22) 5. 4/5 Missed requirements... 6. 4/4 Correct annual savings... 7. 2/3 Used a size weight for cost...
Assaley, Emily	—	163.5 PM	8.1 Months	\$1,962,429		...	18.0	Emily Assaley (18/22) 5. 4/5 Missed requirements... 6. 4/4 Correct annual savings... 7. 3/3 Correct reuse model...
Astrup, Dylan	See attached spreadsheet	163.5 PM	8.1 Months	\$1,962,429	20.2 People	...	19.0	Dylan Astrup (19/22) 5. 4/5 Missed requirements... 6. 4/4 Correct annual savings... 7. 2/3 Used a size weight for cost...
Bernstein, Nicholas	—	145.1 PM	7.8 Months	\$1,741,761	—	...	20.0	Nicholas Bernstein (20/22) 5. 5/5 Correct revised COSYSMO... 6. 4/4 Correct annual savings... 7. 2/3 Used a size weight for cost...

Automated Grader: Early Operating Experience

- Each assignment has required substantial instruction tailoring because problem types, grading criteria, artifacts, and expected solution forms vary widely.
- Revamping the grader instructions for every assignment has been one of the largest setup costs.
- Going forward, the grader is being reorganized around reusable instruction components with transclusion, so common grading policies, output formats, and review rules do not have to be rewritten.
- AI grading has always required substantial human review in early operations to verify, correct, and calibrate scores and comments.
- The workflow is therefore semi-automated: AI drafts grades and feedback, while the instructor remains responsible for final grades.
- Across several thousand graded problems, no students have

figures/bootstra

- Meta-assistant for creating governed academic AI assistants more consistently.
- Generates draft role definitions, instruction sets, knowledge-base structures, output schemas, and test plans.
- Builds from lessons learned in HEATHER 2, Capstone/Thesis Reviewer, and Automatic Grader.
- Reduces ad hoc prompt writing and supports

HEATHER 2 CCB: AI Application Governance Functions

- **Governs HEATHER 2 as an academic AI application**
 - Maintains HEATHER 2 as a controlled, auditable, human-supervised AI assistant.
 - Ensures the AI remains aligned with approved NPS, SE Department, DL, sponsor, and academic procedures.
 - Preserves human authority for admissions, funding, degree eligibility, graduation, exceptions, and policy decisions.
- **Controls AI knowledge and behavior**
 - Reviews proposed changes to SOPs, cheat sheets, policies, knowledge-base files, and GPT instructions.
 - Confirms that HEATHER 2 responses are grounded in authoritative sources rather than informal practice or unsupported inference.
 - Defines what HEATHER 2 may answer directly, what it must caveat,

Human Authority and Student Awareness

- AI outputs are draft decision-support products, not final academic or administrative decisions.
- Faculty and staff review, edit, accept, reject, or override assistant outputs.
- For grading applications, students are informed through course policy language when AI is used to support assessment or feedback.
- The intended use is consistency, coverage, and timely feedback, not replacement of instructor judgment.
- Student reactions and perceived fairness are being captured as part of the grading and assessment pilots.

Governance and Sustainment

- Instructions and knowledge bases are treated as controlled engineering artifacts, not informal prompts.
- HEATHER 2 updates are managed through a formal configuration management board.
- Assistant baselines include intended users, authoritative sources, prohibited behaviors, output formats, and test cases.
- Sustainment uses a DevOps continuous process: collect issues, revise sources, test behavior, approve updates, and archive baselines.
- This reduces risk as staff change, procedures evolve, and model capabilities improve over time.

HEATHER 2 CCB: What It Does for AI Application Governance

Purpose: The HEATHER 2 Configuration Control Board (CCB) governs how the AI assistant changes, learns from approved knowledge sources, and remains trustworthy for NPS academic workflows.

Core CCB Functions

- Reviews and approves changes to HEATHER 2 knowledge bases, GPT instructions, SOP-linked behaviors, and test cases.
- Confirms that proposed AI behavior is grounded in authoritative policy, approved procedures, or designated human expertise.
- Screens proposed updates for PII, sensitive student information, sponsor-sensitive content, and inappropriate data exposure.

Avoiding Over-Constrained AI Behavior

- SOPs and rubrics constrain routine process support, but should not suppress expert judgment or novel student ideas.
- Assistants are instructed to identify uncertainty, ambiguity, and exception cases rather than force mechanical answers.
- Human reviewers remain responsible for recognizing original, unconventional, or high-risk technical approaches.
- Reviews separate compliance concerns from technical originality and mission value.
- Monitoring focuses on both false confidence and excessive rigidity.

Assessment Approach

- Compare AI outputs with human baselines while recognizing variability among human reviewers.
- Use paired statistical tests, effect sizes, correlation analysis, and qualitative faculty review where appropriate.
- Clarify sample sizes and treat early statistical results as calibration evidence, not final proof.
- Calibrate against verified golden examples rather than optimizing only to historical advisor scores.
- Track review time, rework, consistency, completeness, student/faculty trust, and downstream assessment value.

Validation Beyond the SE Department

- The immediate implementation is local to the SE Department.
- Transferability depends on replacing SE-specific SOPs, rubrics, actors, and outputs with those of the target organization.
- The reusable pattern is: define process boundaries, curate authoritative sources, write controlled instructions, test against representative cases, and maintain baselines.
- Cross-context validation will compare accuracy, usefulness, workload reduction, and user trust in each new setting.
- The Bootstrapper captures this pattern so future assistants are engineered rather than improvised.

Integrated Lessons

- The main value comes from combining institutional grounding, constrained behavior, and human review.
- SOPs, rubrics, and assessment instruments become executable knowledge sources when paired with careful instructions.
- Structured outputs connect individual assistance to program assessment and continuous improvement.
- Bootstrapping creates leverage: one well-governed assistant can help produce the next one.
- Trust depends on disclosure, monitoring, configuration control, and clear limits on AI authority.

Conclusions

- HEATHER 2, the Capstone/Thesis Reviewer, Automatic Grader, and AI Bootstrapper form a coherent SE academic AI ecosystem.
- The common pattern is governed decision support for consequential academic processes.
- Current activities are strengthening configuration control, empirical evaluation, rubric calibration, student disclosure, and secure deployment planning.
- The integrated effort improves process continuity, assessment evidence, instructional feedback, and faculty/staff efficiency.
- The SE Department is demonstrating an early adopter path that can later be validated and adapted across campus.